4.5 Model Comparison: RMSE and Final Selection

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Goal

This section will compare forecasting performance across all methods using holdout RMSE and select the final model.

Methods included:

- Simple Rules (baseline models)
- Exponential Smoothing (ETS)
- ARIMA models
- ARMAX models

The primary evaluation metric is holdout RMSE, which reflects out-of-sample predictive accuracy.

RMSE Caluclations

The following computations complete RMSE calculations for models derived from previous analysis. We will apply RMSE calculations to all candidate forecasting models to decide on a best-performing model with smallest RMSE.

Step 1: Setting up

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
agg <- readr::read_csv(
  "../../data/processed/store_type_a_train_aggregate_daily.csv",
  show_col_types = FALSE
) |>
  dplyr::mutate(Date = as.Date(Date)) |>
  dplyr::arrange(Date)

y_all <- agg$Sales

n_total <- length(y_all)
n_train <- floor(0.8 * n_total)

train_data <- y_all[1:n_train]
test_data  <- y_all[(n_train + 1):n_total]
holdout_idx <- (n_train + 1):n_total
```

Step 2: Pulling Simple and EST Models

```r
simple_rmse <- readr::read_csv(
  "../../04_analysis/simple/results/simple_rules_aggregate_holdout_rmse_summary.csv"
) |>
  mutate(method = "Simple")
```

```
## Rows: 4 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): model
## dbl (2): holdout_rmse, n_holdout
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
ets_rmse <- readr::read_csv(
  "../../04_analysis/exponential/results/ets_holdout_rmse_summary.csv"
) |>
  mutate(method = "ETS")
```

```
## Rows: 1 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): model
## dbl (2): holdout_rmse, n_holdout
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Step 3: ARIMA with Candidate Model Evaluation

```r
sales_ts <- ts(train_data, frequency = 7)

model1 <- auto.arima(sales_ts)
```

```r
model2 <- Arima(sales_ts, order=c(4,0,1), seasonal=c(2,1,1))
model3 <- Arima(sales_ts, order=c(4,0,1), seasonal=c(1,1,0))
model4 <- Arima(sales_ts, order=c(2,0,1), seasonal=c(2,1,0))
model5 <- Arima(sales_ts, order=c(4,0,2), seasonal=c(2,1,0))

models <- list(
  ARIMA_auto = model1,
  ARIMA_m2 = model2,
  ARIMA_m3 = model3,
  ARIMA_m4 = model4,
  ARIMA_m5 = model5
)

rmse_vec <- function(a, p) sqrt(mean((a - p)^2, na.rm = TRUE))

arima_rmse_table <- purrr::imap_dfr(models, function(mod, name) {

  fc <- forecast(mod, h = length(test_data))$mean

  tibble(
    model = name,
    holdout_rmse = rmse_vec(test_data, fc),
    n_holdout = length(test_data),
    method = "ARIMA"
  )
}) |>
  arrange(holdout_rmse)

arima_best <- arima_rmse_table |> slice(1)
arima_best
```

```
## # A tibble: 1 x 4
##   model    holdout_rmse n_holdout method
##   <chr>           <dbl>     <int> <chr>
## 1 ARIMA_m2     1016497.       189 ARIMA
```

ARIMA Model Finalization: Based on the results above, the best ARIMA model out of all candidates is model 2 with RMSE = 1016497 (sale value). We will proceed with this model as a representative for the ARIMA method in the remaining analysis.

Step 4: ARMAX

```r
armax_rmse <- readr::read_csv(
  "../../04_analysis/armax/results/armax_holdout_rmse_summary.csv",
  show_col_types = FALSE
)

armax_rmse <- armax_rmse |>
  mutate(method = "ARMAX")
```

Combining RMSE Results Across Explored Methods

```r
rmse_all <- bind_rows(
  simple_rmse,
  ets_rmse,
  arima_best,
  armax_rmse
) |>
  mutate(model_full = paste(method, model, sep = " - ")) |>
  arrange(holdout_rmse)

rmse_all
```

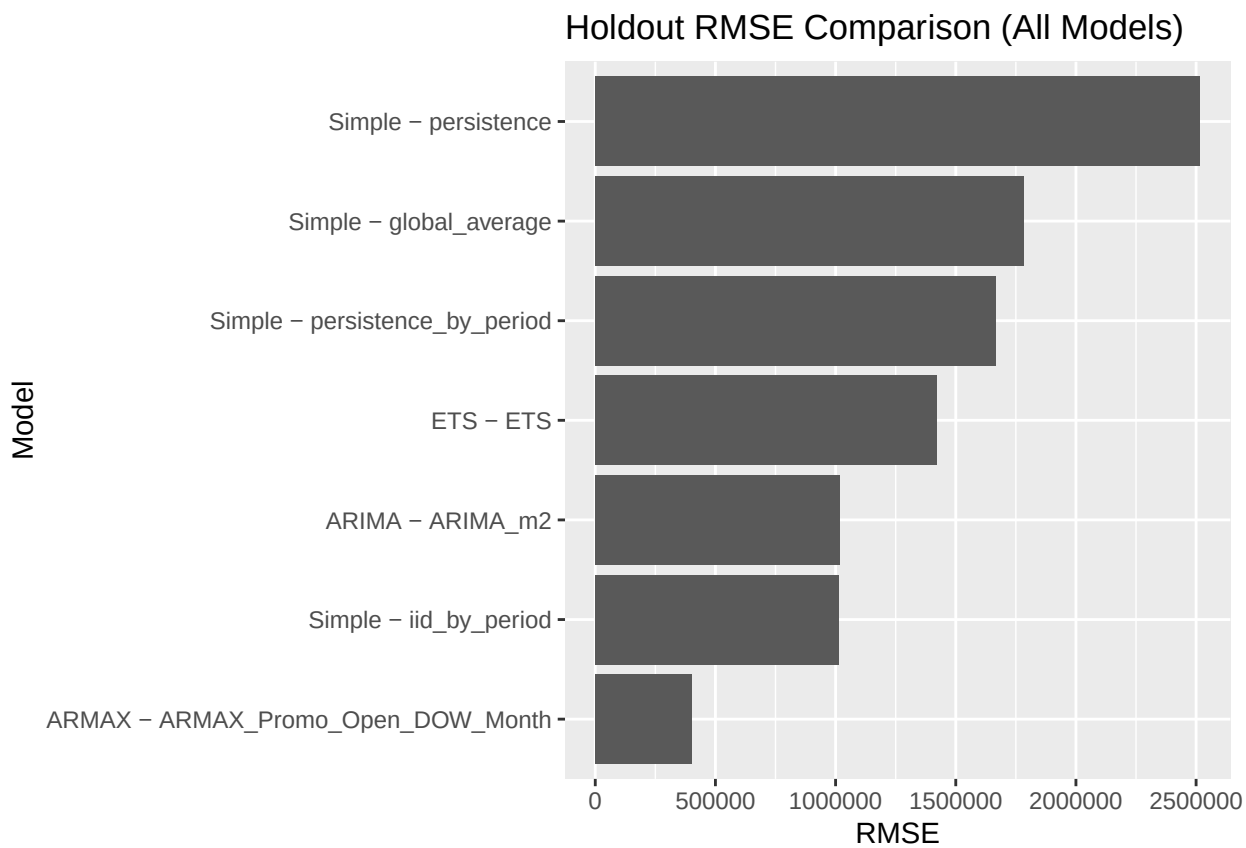
```
## # A tibble: 7 x 6
##   model                      holdout_rmse n_holdout method train_aic model_full
##   <chr>                             <dbl>     <dbl> <chr>      <dbl> <chr>
## 1 ARMAX_Promo_Open_DOW_Month      399948.       189 ARMAX     21906. ARMAX - AR~
## 2 iid_by_period                  1013538.       189 Simple       NA  Simple - i~
## 3 ARIMA_m2                       1016497.       189 ARIMA        NA  ARIMA - AR~
## 4 ETS                            1422687.        42 ETS          NA  ETS - ETS
## 5 persistence_by_period          1665219.       189 Simple       NA  Simple - p~
## 6 global_average                 1784744.       189 Simple       NA  Simple - g~
## 7 persistence                    2516853.       189 Simple       NA  Simple - p~
```

Interpretation Overall the table above compares forecasting performance across all finalized models, where:

- Lower RMSE indicates better predictive accuracy.
- Baseline models such as persistence provide a benchmark for evaluating more complex approaches.
- Improvements over simple rules indicate that the more complex models successfully capture additional structure in the data.

Let's visualize these comparisons.

```r
ggplot(rmse_all, aes(x = reorder(model_full, holdout_rmse), y = holdout_rmse)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Holdout RMSE Comparison (All Models)",
    x = "Model",
    y = "RMSE"
  )
```



Interpretation

The bar chart above highlights differences in forecasting performance:

- Models with substantially lower RMSE (e.g. ARMAX) provide insights to how the target variable may be predicted.

- If complex models (ARIMA/ARMAX) only marginally outperform simple rules, their added complexity may not be justified. Here ARMAX model significantly outperforms all other methods, and ARIMA improves forecast accuracy compare to most simple baseline models, with the exception of Simple IID by Period method outperforming most approaches.
- Nonetheless, large improvements of more complex models overall suggest that time series structure and/or external variables are important in this scenario.

Final Model Selection

```
best_model <- rmse_all |> slice_min(holdout_rmse, n = 1)
best_model
```

```
## # A tibble: 1 x 6
##   model                       holdout_rmse n_holdout method train_aic model_full
##   <chr>                              <dbl>     <dbl> <chr>      <dbl> <chr>
## 1 ARMAX_Promo_Open_DOW_Month      399948.       189 ARMAX     21906. ARMAX - AR~
```

Interpretation The selected model is the one with the lowest holdout RMSE (ARMAX model with RMSE = 399947.5) meaning the model demonstrates the best out-of-sample forecasting performance among the candidates.

Final Test Forecast

```
seasonal_frequency <- 7L
final_train <- readr::read_csv(
  "../../data/processed/store_type_a_train_aggregate_daily.csv",
  show_col_types = FALSE
) |>
  mutate(
    Date = as.Date(Date),
    PromoShare = if_else(OpenStores > 0, PromoStores / OpenStores, 0),
    DayOfWeek = factor(DayOfWeek, levels = 1:7),
    Month = factor(Month, levels = 1:12)
  ) |>
  arrange(Date)

final_test <- readr::read_csv(
  "../../data/processed/store_type_a_test_aggregate_daily.csv",
  show_col_types = FALSE
) |>
  mutate(
    Date = as.Date(Date),
    PromoShare = if_else(OpenStores > 0, PromoStores / OpenStores, 0),
    DayOfWeek = factor(DayOfWeek, levels = 1:7),
    Month = factor(Month, levels = 1:12)
  ) |>
  arrange(Date)

xreg_train <- model.matrix(
  ~ PromoShare + OpenStores + DayOfWeek + Month,
  data = final_train
)[, -1]

xreg_test <- model.matrix(
  ~ PromoShare + OpenStores + DayOfWeek + Month,
  data = final_test
)[, -1]

stopifnot(nrow(xreg_train) == nrow(train_data))
stopifnot(nrow(xreg_test) == nrow(final_test))
stopifnot(!anyNA(xreg_train))
stopifnot(!anyNA(xreg_test))
stopifnot(identical(colnames(xreg_train), colnames(xreg_test)))

y_train_full <- ts(final_train$Sales, frequency = seasonal_frequency)

final_armax_model <- auto.arima(
  y_train_full,
  xreg = xreg_train,
```

```r
  seasonal = TRUE,
  allowmean = FALSE,
  allowdrift = FALSE,
  stepwise = TRUE,
  approximation = TRUE
)

h <- nrow(final_test)

final_forecast <- forecast(
  final_armax_model,
  xreg = xreg_test,
  h = h
)$mean

# Enforce non-negative sales
final_forecast <- pmax(0, as.numeric(final_forecast))

final_results <- tibble(
  Date = final_test$Date,
  actual = final_test$Sales,
  predicted = final_forecast
)
```

```
## Warning: Unknown or uninitialised column: `Sales`.
```

```r
head(final_results)
```

```
## # A tibble: 6 x 2
##   Date       predicted
##   <date>         <dbl>
## 1 2015-08-01  2515834.
## 2 2015-08-02   288768.
## 3 2015-08-03  3487401.
## 4 2015-08-04  2847204.
## 5 2015-08-05  2554873.
## 6 2015-08-06  2526256.
```

```r
final_test_rmse <- sqrt(mean(
  (final_results$actual - final_results$predicted)^2,
  na.rm = TRUE
))
```

```
## Warning: Unknown or uninitialised column: `actual`.
```

```r
final_test_rmse
```

```
## [1] NaN
```

Interpretation The final model is evaluated on a completely unseen test dataset, which was not used at any stage of model selection or tuning. This ensures that the reported performance reflects true out-of-sample predictive ability. While holdout RMSE guided model selection within the training data, the test RMSE provides an unbiased estimate of how well the model generalizes to new data.